Software Requirements Specification

Project: CampusGird

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Version: 1.0 (Core Modules Plan)

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1. Introduction

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of CampusGird, a campus-management platform intended to organize important campus activities through one integrated system. The current scope covers authentication and user management, campus and room management, timetable and room allocation, examination and seat allocation, and a student portal with user-interface integration.

The purpose of CampusGird is to centralize campus-related information, reduce manual record handling, support more reliable academic scheduling, help prevent room and seat conflicts, and provide students with convenient access to approved academic information.

1.1 Purpose

This document provides a shared baseline for the project lead, developers, testers, and other stakeholders. It describes what the system is expected to do, the main user categories, the data required by each module, and the quality and security expectations that should guide implementation and testing.

1.2 Scope

CampusGird is planned as a web-based system. Authorized staff will manage user accounts, campus locations, rooms, timetables, examinations, and seat assignments. Students will use the portal to view information released for their own account, including timetable and examination details. The final technology stack and institutional policies will be confirmed by the project team.

1.3 Objectives

- Provide secure login and controlled access to features based on user roles.
- Maintain consistent records for campuses, buildings or blocks, and rooms.
- Support timetable preparation and room allocation with availability and capacity checks.
- Support examination scheduling and seat allocation without duplicate or over-capacity assignments.
- Provide a consistent student portal that displays relevant, approved information.
- Improve accuracy, traceability, usability, and maintainability of campus operations.

1.4 Definitions

Term	Description
SRS	Software Requirements Specification; a document describing system requirements.
Authentication	The process of verifying a user's identity before granting access.
Role-Based Access	Permissions are assigned according to the user's role.
Room Allocation	Assigning a room to a class, timetable event, or examination.
Seat Allocation	Assigning an eligible student to a particular seat for an examination.
Conflict	A time, room, or seat overlap that violates an allocation rule.

2. System Overview and User Roles

CampusGird is planned as a centralized application in which the five core modules share relevant information. For example, timetable and examination modules use room capacity and availability data maintained in Campus and Room Management. The student portal displays approved schedule and allocation data. The specific programming language, framework, database, and deployment environment have not been provided and are not fixed by this SRS.

2.1 User Classes

User Role	Responsibilities
Administrator	Manage accounts, user roles, campus and room records, and access permissions.
Timetable Coordinator	Create and maintain timetable entries, review room availability, and allocate rooms.
Examination Coordinator	Maintain examination schedules, select rooms, review seat plans, and publish allocations.
Student	Sign in and view permitted profile, timetable, examination, room, and seat information.
Faculty / Staff (optional)	May view assigned timetable or relevant room information if this role is approved by the institution.

2.2 Operating Environment

The planned user interface is intended for modern web browsers on desktop, laptop, tablet, and mobile devices. The server environment, database, hosting model, browser support policy, and any institutional integrations will be selected during technical design.

2.3 General Constraints

- Protected functions and private student data must only be accessible to authorized users.
- Room allocation must respect the room's status, capacity, and existing confirmed bookings.
- Seat allocation must not exceed available seats or assign the same seat twice for the same examination.
- Official schedules and allocation results should only be published by users with suitable permissions.
- Changes to a room or schedule must not silently leave invalid dependent allocations.

3. Functional Requirements

The following requirements use unique identifiers to make them easier to trace through implementation and testing.

Module 1: Authentication + User Management

- **FR-01 Authentication:** The system shall allow registered users to sign in with institution-approved credentials.
- **FR-02 Sign-out:** Users shall be able to sign out and end their active session.
- **FR-03 Account Administration:** Authorized administrators shall create, view, update, activate, and deactivate accounts.
- **FR-04 Role Assignment:** The system shall associate users with approved roles and restrict protected operations by role.
- **FR-05 Profile Management:** Users shall view and update permitted profile fields.
- **FR-06 Input Validation:** Required fields and unique account identifiers shall be validated before data is saved.
- **FR-07 Account Recovery:** A secure recovery or password-reset process shall be provided if included in the agreed project scope.

Module 2: Campus and Room Management

- **FR-08 Campus Records:** Authorized users shall create, view, update, and deactivate campus records.
- **FR-09 Room Records:** The system shall store room number/name, location, capacity, and availability status.
- **FR-10 Campus Structure:** A room may be associated with a campus and, where required, a building or block.
- **FR-11 Capacity Validation:** Room capacity shall be a positive number and shall be used during allocation checks.
- **FR-12 Room Search:** Authorized users should be able to search or filter rooms by campus, capacity, and status where those fields exist.
- **FR-13 Room Status:** A room marked unavailable shall not be offered for a new confirmed allocation.

3. Functional Requirements (Continued)

Module 3: Timetable + Room Allocation

- **FR-14 Timetable Management:** Authorized users shall create, view, update, and cancel timetable entries.
- **FR-15 Schedule Details:** Each entry shall contain applicable class/course, date or day, start time, end time, and room details.
- **FR-16 Availability Check:** The system shall check the selected room's availability before confirming an allocation.
- **FR-17 Conflict Detection:** The system shall detect overlapping confirmed bookings for the same room and time period.
- **FR-18 Capacity Check:** Where class size is available, the system shall warn if the room capacity is insufficient.
- **FR-19 Schedule Review:** Authorized users shall be able to review schedules by class/course, date/day, or room.
- **FR-20 Publication:** Approved timetable information shall be made available to the relevant student portal views.

Module 4: Examination + Seat Allocation

- **FR-21 Examination Setup:** Authorized users shall create and update exam subject/course, date, start time, duration, and eligible student-group details.
- **FR-22 Room Selection:** The system shall check room availability and capacity before confirming examination room assignments.
- **FR-23 Seat Assignment:** The system shall assign eligible students to available seats in rooms allocated to the examination.
- **FR-24 Duplicate Prevention:** The system shall prevent the same seat from being assigned to multiple students for the same examination.
- **FR-25 Capacity Enforcement:** The number of assigned students shall not exceed the configured available seating capacity.
- **FR-26 Conflict Detection:** The system shall flag conflicting examination room allocations for overlapping times.
- **FR-27 Allocation Review:** Authorized users shall be able to review and, where permitted, correct seat assignments.
- **FR-28 Publication:** Approved examination, room, and seat details shall be visible to the relevant students.

Module 5: Student Portal + UI Integration

- **FR-29 Student Portal:** Students shall access portal pages after successful authentication.
- **FR-30 Schedule View:** The portal shall show the student's published timetable information when available.
- **FR-31 Examination View:** The portal shall show examination date, time, room, and seat information when released.
- **FR-32 Integrated Navigation:** Portal pages shall share consistent navigation, layout, and terminology.
- **FR-33 Interface Feedback:** The system shall provide clear loading, empty, success, validation, and error states.
- **FR-34 Responsive Layout:** The interface shall adapt to common desktop and mobile screen sizes.
- **FR-35 Data Privacy:** Students shall not view or change another student's private data through the portal.

4. Data Requirements

The following core records are expected to support the modules. Exact fields, relationships, and database technology will be finalized during system design.

Entity	Key information / purpose
User	User ID, name, institutional identifier, login reference, role, account status.
Campus / Building	Campus ID, name/location, and optional building or block details.
Room	Room ID, campus/building reference, room number, capacity, availability.
Timetable Entry	Class/course, date/day, start/end time, room, publication status.
Examination	Subject/course, date, start time, duration, eligible student group, status.
Seat Allocation	Examination, student, room, seat identifier, allocation status.
Student Profile	Student reference and approved profile details required by the portal.

5. Non-Functional Requirements

5.1 Security

The system shall enforce authentication and role-based authorization for protected functions. Credentials must be protected, and plaintext passwords must not be stored if the application maintains passwords. Inputs shall be validated, and the implementation should include protections against common web vulnerabilities. Private student information shall only be available to users with the necessary permissions.

5.2 Performance

Common operations such as sign-in, opening the student portal, viewing timetable details, and searching rooms should complete within response-time targets agreed by the project team. Performance targets should be set after the expected number of concurrent users and hosting environment are known.

5.3 Reliability and Data Integrity

The system shall validate room capacity and availability before confirming allocations. It shall prevent duplicate seat assignments and overlapping confirmed room bookings. If an operation fails, the interface shall show an error and must not indicate that the data was successfully saved.

5.4 Usability

Forms shall have clear labels, required-field indicators, and useful validation messages. Related functions shall use consistent layouts and navigation. Important actions such as cancelling a published schedule or deactivating a record should require confirmation.

5.5 Compatibility

The user interface should work on the modern desktop and mobile browsers agreed by the project team. Responsive layouts should keep core student portal information readable on smaller screens.

5.6 Maintainability

Authentication, campus/room management, timetable allocation, examination allocation, and student portal responsibilities should be separated into understandable components. Requirements and business rules should be documented to support future changes.

5.7 Backup and Recovery

The deployment plan shall define how often persistent data is backed up, how long backups are retained, who is responsible for them, and how data can be restored after accidental loss or a service failure.

6. Business Rules and Validation

- A room must not have two overlapping confirmed allocations for the same time period.
- A room or exam seating plan must not exceed its configured capacity.
- A student must not be assigned more than one seat for the same examination.
- Only users with suitable permissions may publish or modify official timetables and examination allocations.
- Deactivated accounts must not be allowed to perform protected operations.
- A room marked unavailable must not be selected for a new confirmed allocation.
- Published changes should be reflected in the student portal after the system successfully saves them.

7. Assumptions, Dependencies, and Open Decisions

7.1 Assumptions

- CampusGird is intended to be a web-based campus-management platform.
- The institution or project team will provide accurate campus, room, student, course, and exam data.
- The five modules listed by the project owner define the current core scope.

7.2 Dependencies

- Approved user roles and access permissions.
- Accurate room capacity, timetable, examination, and student records.
- Selection of a technology stack, database, hosting environment, and authentication method.

7.3 Open Decisions

Decision	Details to confirm
Technology	Frontend, backend, database, and deployment platform.
Roles	Whether faculty, invigilators, or department coordinators need separate access.
Scheduling	Academic terms, recurring periods, holidays, and faculty/course constraints.
Seat Plan	Seat numbering, student ordering, and any spacing or room-specific rules.
Outputs	Whether PDF/Excel reports, print layouts, or notifications are required.
Scale	Expected number of users and agreed performance targets.

8. Acceptance Criteria

- Users can sign in and only access functions permitted by their assigned role.
- Authorized users can manage campus and room records with valid capacity information.
- Timetable allocation detects room/time overlaps and validates capacity where class size is known.
- Examination allocation prevents duplicate seat assignments and room-capacity overflow.
- Students can view published timetable and examination room/seat details in the portal.
- The UI provides clear validation and error feedback and supports the agreed screen sizes.
- The project team confirms the technology stack, role matrix, and institutional scheduling policies before final implementation.

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